CAI BIRCH (HND/ B.Sc./ M.Sc.)

Software Developer – backend, integrations and secure systems

Stockholm | 076-185 95 27 | caiowain@gmail.com | www.caibirch.com | github.com/CadfanCode

PROFILE

Software developer specialising in backend development with Java, Kotlin and Spring Boot. Has built and deployed cloud-based microservices in agile teams using PostgreSQL, Hibernate/JPA, REST APIs, Docker and Kubernetes, and has also worked with Python and AWS through studies and personal projects. Experienced with business-critical flows for sensitive data and real-time communication, where performance, stability and availability are key. A background as a researcher and university lecturer gives strong analytical skills for solving complex problems in a structured way and explaining technical solutions clearly. Thrives in teams with a high degree of ownership, following the principle of ”you build it, you own it”.

TECHNICAL SKILLS

Languages	Java, Kotlin, Python, JavaScript, TypeScript, SQL, C#
Backend & data	Spring Boot (v2/v3), REST APIs, microservices, PostgreSQL, Hibernate/JPA, WebRTC, JSON, XML
Cloud & DevOps	AWS, Docker, Kubernetes, CI/CD, Git, GitHub, JUnit, Mockito, JIRA
Frontend & web	React, React Native, Vue.js, Expo, HTML, CSS, Three.js
Methodology	OOP, data structures and algorithms, design patterns, unit testing, secure development, Agile/Scrum, Claude Code.
Spoken languages	Swedish and English – fluent, spoken and written

WORK EXPERIENCE

Software Developer (Internship) | Braive AB

Stockholm | 2026

- Developed and deployed microservices in Java/Kotlin with Spring Boot as well as admin interfaces in Vue.js, in an agile team using Git, JIRA and code reviews.
- Improved the stability of the backend's logging flows by identifying and resolving race conditions, reducing operational disruptions and strengthening data integrity.
- Optimised the handling of sensitive healthcare data in PostgreSQL and Hibernate/JPA based on IT security principles, reducing sources of manual error.
- Containerised services with Docker and worked in Kubernetes-based DevOps pipelines for consistent environments from development to production.

Software Developer (Internship) | Omika Health AB

Stockholm | 2025

- Designed and implemented a production-ready P2P video chat with WebRTC in React Native, including a lightweight signalling server and secure authentication.
- Built and debugged low-latency real-time flows and verified complex client interactions in proof-of-concept tests.
- Turned user needs into concrete technical solutions, working closely with a small, product-focused team.

Correctional Officer (hourly and summer cover) | Swedish Prison and Probation Service

Sweden | Ongoing

- Works in an environment with high demands on security, confidentiality, stress tolerance and sound judgement.
- Brings an operational, real-world perspective into software development by linking technical solutions to actual user needs and security requirements.

University Lecturer, laboratory teaching | Loughborough University

Loughborough, United Kingdom | 2018–2021

- Responsible for advanced-level teaching in biomechanics and motor control for over 100 students, using Matlab and data collection protocols in OpenSim.
- Mentored colleagues in interdisciplinary teams and drove a knowledge-sharing, collaborative way of working.

Research Assistant | GIH / KTH
Stockholm | 2017

- Analysed biomechanical data using quantitative modelling to identify patterns of muscle spasticity in stroke patients.
- Applied analytical problem-solving and data interpretation in a research setting with complex movement data.

EDUCATION

Higher Vocational Diploma in Java Software Development | YH Akademin, Sundsvall
2024–2026
Two-year higher vocational programme with a clear focus on backend development in Java and practical, industry-oriented application through two internship periods at Braive AB and Omika Health AB.

- Java and object-oriented programming: data structures, algorithms, design patterns and unit testing with JUnit and Mockito.
- Backend development with Spring Boot: REST APIs, microservice architecture, Hibernate/JPA as well as relational databases and SQL.
- System architecture and scalable solutions: database management, integrations and secure development.
- Modern cloud infrastructure and DevOps: Docker, CI/CD and version control with Git in team-based, agile projects following Scrum.
- Supplementary modules in Python as well as web development with JavaScript, HTML and CSS.

Neuroscience (doctoral studies) | Lund University
Doctoral studies in neuroscience with a focus on motor control. Advanced quantitative data analysis, experimental design and scientific writing – training in breaking down complex problems, working hypothesis-driven and drawing conclusions from large datasets.

Master's degree, Sport Science | GIH, Stockholm
Specialisation in biomechanics and movement analysis. Extensive data collection and signal processing of movement data as well as statistical analysis and modelling in Matlab.

Bachelor's degree, Sport Science | Bangor University, United Kingdom
Broad foundation in research methodology, statistics and critical thinking, with training in planning studies, collecting and interpreting data, and presenting results clearly both orally and in writing.

Associate Fellow of Advance HE (teaching accreditation) | Loughborough University
Internationally recognised teaching accreditation in higher education – evidence of the ability to explain complex technical subjects clearly and adapt communication to the audience.

GITHUB PROJECTS

All projects are available at **github.com/CadfanCode** – here is a selection:

- **Portfolio with the AI parrot Polly** (www.caibirch.com) – Interactive website built with Three.js where visitors can talk code with Polly, a lightweight AI assistant powered by a local LLM model. My playground for testing new technologies and ideas.
- **SchoolProject_API** (Java) – REST API built with Spring Boot and Spring Data JPA, focusing on clean architecture, entity modelling and database management.
- **SchoolProject_InterviewDemoForInternship** (Java) – API demo developed ahead of a technical interview, showing how I structure and present code under time pressure.
- **Wedding2026** (HTML/CSS/JavaScript) – Wedding website in live production with real users, from design to publication.
- **Personal ML project** – Developing a tool that uses machine learning to analyse kinematic data in order to identify risk factors for musculoskeletal injuries.

ACHIEVEMENTS

- **3rd place, Innovation Pioneers Hackathon 2025** – Prototyped, in 48 hours, an AI-driven collaboration app connecting entrepreneurs with key stakeholders; top 3 in a competition judged by innovation experts.
- **Jfokus 2026** – Attended one of Europe's leading developer conferences, focusing on modern microservice architectures, tooling and security trends in the Java ecosystem.

REFERENCES

Available on request